# Curriculum Vitae

## Personal Information


Hongyu Li
Brown University
Department of Computer Science
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## Education

10/2025–Current **Visiting Student**
Columbia University, New York, NY, USA
**Advisor:** Professor Yunzhu Li

08/2023–Current **Ph.D., Computer Science**
Brown University, Providence, RI, USA
**Advisor:** Professor George Konidaris

08/2021–05/2023 **M.Sc., Computer Science**
Northeastern University, Boston, MA, USA
**Thesis:** Toward Stereo-based Obstacle Detection using Efficient Deep Neural Network
**Advisor:** Professor Taşkın Padır
**Advisor:** Professor Huaizu Jiang

08/2018–12/2020 **B.Sc., Computer Science**
Rensselaer Polytechnic Institute, Troy, NY, USA
**Second Major:** Economics
**Honors:** Cum Laude, Dean's Honor List

## Experience

06/2025–Current **Research Intern**
Robotics and AI Institute (fka Boston Dynamics AI Institute), Cambridge, MA

Mentors: Dr. Jiahui Fu and Dr. Lingfeng Sun.
Manipulation using video generative models.

05/2024–09/2024 **Applied Scientist II Intern**
Amazon Robotics (Innovation Lab), Westborough, MA
Hosts: Professor Taşkın Padır and Tye Brady.
Researched in-hand object 6D pose tracking using visuotactile perception. Published paper **C.4**.

05/2023–08/2023 **Research Intern**
Honda Research Institute, San Jose, CA
Hosts: Dr. Nawid Jamali and Dr. Soshi Iba.
Researched visuotactile perception. Published paper **C.7**. Filed patents **P.1** and **P.2**.

09/2022–12/2022 **Research Intern**
Honda Research Institute, San Jose, CA
Hosts: Dr. Nawid Jamali and Dr. Soshi Iba.
Researched in-hand object 6D pose estimation using visuotactile perception. Published paper **J.2**. Filed patents **P.3**.

09/2021–05/2023 **Graduate Research Assistant**
Robotics and Intelligent Vehicles Research Lab, Northeastern University, Boston, MA
PI: Professor Taşkın Padır
Researched in the fields of robot perception and navigation. Published papers **J.1**, **C.8**, and **C.9**.

03/2021–06/2021 **Artificial Intelligence Intern**
KPMG, Nanjing, China
Developed the speech processing modules for the KPMG AI Factory Platform.

## Publications

$*$ or $^{\dagger}$ represents equal contribution or equal advising, depending on the roles of the authors.

### Journals

**J.1** Linfeng Zhao*, **Hongyu Li***, Taşkın Padır, Huaizu Jiang$^{\dagger}$, Lawson L.S Wong$^{\dagger}$. *E(2)-Equivariant Graph Planning for Navigation.* Accepted by IEEE Robotics and Automation Letters (RA-L) vol. 9, no. 4. Present at IROS 2024.

**J.2** **Hongyu Li**, Snehal Dikhale, Soshi Iba, Nawid Jamali. *ViHOPE: Visuotactile In-Hand Object 6D Pose Estimation with Shape Completion.* Accepted by IEEE Robotics and Automation Letters (RA-L) vol. 8, no. 11. Present at ICRA 2024.

### Conferences

**C.1** **Hongyu Li***, Lingfeng Sun, Yafei Hu, Duy Ta, Jennifer Barry, George Konidaris, Jiahui Fu. *NovaFlow: Zero-Shot Manipulation via Actionable Flow from Generated Videos.* Under review.

**C.2** Wanjia Fu*, **Hongyu Li***, Ivy X. He, Stefanie Tellex, and Srinath Sridhar. *UniTac: Whole-Robot Touch Sensing Without Tactile Sensors.* Under review.

**C.3** **Hongyu Li**, Mingxi Jia, Mete Tuluhan Akbulut, Yu Xiang, George Konidaris, Srinath Sridhar. *V-HOP: Visuo-Haptic 6D Object Pose Tracking.* Accepted by Robotics: Science and System (RSS) 2025.

**C.4** **Hongyu Li**, James Akl, Srinath Sridhar, Tye Brady, Taşkın Padır. *ViTa-Zero: Zero-shot Visuotactile Object 6D Pose Estimation.* Accepted by IEEE International Conference on Robotics and Automation (ICRA) 2025.

**C.5** **Hongyu Li**, Taşkın Padır, Huaizu Jiang. *StereoNavNet: Learning to Navigate using Stereo Cameras with Auxiliary Occupancy Voxels.* Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024.

**C.6** Tianye Ding*, **Hongyu Li***, Huaizu Jiang. *ODTFormer: Efficient Obstacle Detection and Tracking with Stereo Cameras Based on Transformer.* Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024.

**C.7** **Hongyu Li**, Snehal Dikhale, Jinda Cui, Soshi Iba, Nawid Jamali. *HyperTaxel: Hyper-Resolution for Taxel-Based Tactile Signal Through Contrastive Learning.* Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024.

**C.8** **Hongyu Li**, Zhengang Li*, Neşet Ünver Akmandor*, Huaizu Jiang, Yanzhi Wang, Taşkın Padır. *StereoVoxelNet: Real-Time Obstacle Detection Based on Occupancy Voxels From a Stereo Camera Using Deep Neural Networks.* Accepted by IEEE International Conference on Robotics and Automation (ICRA) 2023.

**C.9** Neşet Ünver Akmandor, **Hongyu Li**, Gary M. Lvov, Eric Dusel, Taşkın Padır. *Deep Reinforcement Learning based Robot Navigation in Dynamic Environments using Occupancy Values of Motion Primitives.* Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022.

## Presentations

1. **Hongyu Li**, Mingxi Jia, Mete Tuluhan Akbulut, Yu Xiang, George Konidaris, Srinath Sridhar. *V-HOP: Visuo-Haptic 6D Object Pose Tracking.* Oral Talk at New England Manipulation Symposium (NEMS) 2025.
2. **Hongyu Li**, Snehal Dikhale, Soshi Iba, Nawid Jamali. *ViHOPE: Visuotactile In-Hand Object 6D Pose Estimation with Shape Completion.* Spotlight Talk at NeurIPS 2023 Workshop on Touch Processing.
3. **Hongyu Li**, Huaizu Jiang, Taşkın Padır. *StereoNavNet: Learning to Navigate using Stereo Camera with Auxiliary Occupancy Voxels.* Spotlight Talk at CVPR 2023 Workshop on 3D Vision and Robotics.

4. **Hongyu Li**, Zhengang Li*, Neşet Ünver Akmandor*, Huaizu Jiang, Yanzhi Wang, Taşkın Padır. *StereoVoxelNet: Real-Time Obstacle Detection Based on Occupancy Voxels From a Stereo Camera Using Deep Neural Networks.* Presented at IROS 2022 workshop "Agile Robotics: Perception, Learning, Planning, and Control," Kyoto, Japan, October 2022.

## Patents

**P.1** Nawid Jamali, **Hongyu Li**, Soshi Iba. SYSTEMS AND METHODS FOR TAXEL HYPER-RESOLUTION THROUGH MULTI-CONTACT LOCALIZATION. *Patent Pending.*

**P.2** **Hongyu Li**, Nawid Jamali, Soshi Iba. TAXEL-BASED TACTILE SENSOR. *Patent Pending.*

**P.3** **Hongyu Li**, Nawid Jamali, Snehal Dikhale, Soshi Iba. SYSTEMS AND METHODS FOR VISUOTACTILE OBJECT POSE ESTIMATION WITH SHAPE COMPLETION. US-20240371022-A1. *Patent Pending.*

**P.4** Feiyu Zhu, Tianjian Dai, **Hongyu Li**, Tiancheng Mai, Yuchen Liang, Xiaojing Su. A kind of ball serving device for balls sport training. Granted CN107007996A

## Teaching

01/2024–05/2024 **Graduate Teaching Assistant**
Department of Computer Science, Brown University, Providence, RI
CSCI 1430 Introduction to Computer Vision

09/2021–05/2022 **Graduate Teaching Assistant**
Khoury College of Computer Sciences, Northeastern University, Boston, MA
CS 7610 Foundations of Distributed Systems
DS 2500 Intermediate Programming with Data

09/2019–07/2020 **Teaching Assistant**
Department of Computer Science, Rensselaer Polytechnic Institute, Troy, NY
CSCI 1190 Beginning Programming for Engineers

## Services

### Reviewer

Conferences

1. IEEE International Conference on Robotics and Automation (**ICRA**) 2024, 2025, 2026
2. IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**) 2024, 2025
3. IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) 2024, 2025
4. European Conference on Computer Vision (**ECCV**) 2024
5. International Conference on Computer Vision (**ICCV**) 2025

6. Winter Conference on Applications of Computer Vision (**WACV**) 2026
7. Annual AAAI Conference on Artificial Intelligence (**AAAI**) 2025, 2026
8. ACM/IEEE International Conference on Human-Robot Interaction (**HRI**) 2026
9. ACM CHI Conference on Human Factors in Computing Systems (**CHI**) 2024
10. IEEE-RAS International Conference on Humanoid Robots (**Humanoids**) 2024

Journals

11. IEEE Transactions on Robotics (**T-RO**)
12. IEEE Robotics and Automation Letters (**RA-L**)
13. IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**)
14. IEEE Transactions on Industrial Electronics (**TIE**)
15. Elsevier **Neurocomputing**

Providence, RI, October 8, 2025